

27

COMMUNICATION OF MATCH ANALYSIS

Philip Furley

“Before we demand more of our data, we need to demand more of ourselves” (Silver 2012 p. 9)

OVERVIEW

Learning Goals

- Critically reflect on match analysis data.
- Understand problems in data-based reasoning.
- Being able to name problems in communicating analysis results.
- Reduce Big Data in sports to recommendations for action.
- Understand the influence of tactical instructions on players.

27.1 Theoretical Background

The quote from Nate Silver’s bestselling book *The Signal and the Noise* summarizes the gist of this chapter. Undeniably, the job of scouts and data analysts in sport is not done by simply collecting more and more data. Instead reasonable conclusions must be drawn from these data to inform decision and future courses of action. These conclusions or results of match analyses must then be communicated and passed on to inform training and competition in a targeted manner. Unfortunately, these two aspects (interpretation of match analysis results and communication of match analysis results) are hardly investigated and poorly understood in sport.

This chapter utilizes the example of exploiting match analyses data to prepare the team tactically for the next match. Following from the previous chapter by Wunderlich, the aim of this chapter is to exemplify difficulties that result from the ever-increasing volume of data. More specifically, I show that Big Data in sports can actually be problematic in drawing conclusions and giving data-based recommendations to inform training and competition in sports. As there are hardly any empirical studies on this topic, I will first review related research and then attempt to translate these into evidence-informed recommendations in sports.

27.1.1 Drawing Conclusions Based on Data

In professional sports one can almost speak of an obsession with numbers and statistics. Scientists have even speculated that one reason for the profound gender gap in sports interest is also due to gender-specific predispositions for numbers and statistics (Baumeister 2010). For example, young boys start at early ages to collect and compare sports cards of footballers, baseball players, etc. that contain players' statistics. This fascination with sports stats seems to intensify as they grow up, since adult men frequently discuss sport stats when watching major sports events: e.g., "batting percentage" in baseball, "average yards" in American football, serve strength in tennis or won duels in soccer. It is therefore unsurprising that sports broadcasts in the media respond to this interest by increasingly showing statistics and visualizations of the collected data.

The growing quantification of sport can be partly explained by the fact that people have great confidence in numbers, which is deeply rooted in human psychology (Porter 2020). The "Big Data" movement, which is also becoming increasingly influential in sport (Rein and Memmert 2016), is probably explained, in part, by the human predisposition towards numbers. However, numbers need to be interpreted: "The numbers have no way of speaking for themselves. We speak for them" (Silver 2012, p. 9). In addition, the former editor of the popular *Wired* magazine, Chris Anderson, already noted in 2008 that huge amounts of data can only be interpreted with a meaningful theory and even require the scientific method. These remarks by renowned people in the Big Data field underline a fundamental scientific insight, which is often summarized with a quote from Albert Einstein (cited in Wilson 2019, p. 3): "It is the theory that decides what we can perceive." In other words, it is very unlikely that we will draw important conclusions from ever-increasing amounts of data if we do not have a corresponding theory of how the data should be evaluated or interpreted.

For this reason, science has attempted to develop methods to standardize or objectify the process of drawing conclusions based on data. However, a recently published research project has shown that even science has considerable problems drawing reliable conclusions from a given data set. This research project has become prominent under the title "Many Analysts, One Dataset" (Silberzahn et al. 2018).

EYE CATCHER

Many Analysts, One Dataset; Silberzahn et al. (2018)

In this project, 29 research groups (61 analysts) were given the same soccer data set and were asked to answer the seemingly simple question of whether football referees ($N = 3147$) are more likely to give a red card to players ($N = 2053$) with dark skin color than to players with light skin color. In total, this match analysis data set yielded 146,028 player-referee dyads (the link to the data set can be found here: <https://osf.io/fv8c3/>). The research groups were free to decide how to analyze the data set to answer the question. Remarkably, the research groups came to very different results. 20 research teams (69%) found a statistically significant effect and concluded that dark-skinned players were more likely to receive a red card, whereas 9 teams (31%) came to the opposite conclusion. This result shows that even if experts with years of methodological training analyze and interpret exactly the same data, they cannot answer a relatively simple yes/no question consistently: "significant variation in the results of analyses of complex data may be difficult to avoid, even by experts with honest intentions" (Silberzahn et al. 2018, p. 338).

The project by Silberzahn et al. (2018) should highlight that “data cannot speak for itself”. As coaches and game analysts in sports typically attempt to use more complex match analysis data to inform more complex questions, it seems likely that there would be even less agreement on how to interpret the same data set in this context. It would certainly be interesting to see, for example, how much agreement there would be between different scouts and data analysts from different sports teams on the same match analysis data set. Although it is assumed that match analysis data is objective, it is very unlikely that it will be interpreted and passed on without subjective bias. What makes matters even worse is that, match analysis does not stop at analyzing and interpreting data, but the analysis results are communicated to players and other staff to inform future action. Research suggests that further difficulties can arise here.

27.1.2 “Too Much of a Good Thing”

A much-cited book in the field of decision-making with the title *The Paradox of Choice: Why more is less* by Barry Schwartz (2016) lists a variety of everyday examples, all of which indicate that more information and more decision options do not necessarily make decisions easier, but can actually make them more difficult. Another central message of the book is that people are often unhappier with their decisions when they have a vast amount of decision options or need to integrate a lot of information into their decisions. For example, people become more anxious about making mistakes, about not being able to integrate or weight the multitude of information in a meaningful way, or about the feeling that another option might have been the better one, etc. This phenomenon has become known in psychology as the “too much choice” effect. Although decision researchers conclude that it is not per se better or worse to have many options (e.g., Scheibehne, Greifeneder and Todd 2010), a relevant finding for match analysis in sports shows that the “too much choice” effect depends, among other things, on whether the person making the decision is required to justify their decision (Scheibehne, Greifeneder and Todd 2009). Therefore, it seems likely that coaches or match analysts in sports may encounter similar problems due too much information and decision options as they are usually required to justify their decisions.

Further studies (see Glöckner and Betsch 2012) from the field of cognitive psychology also indicate that the human information processing system has problems to integrate vast amounts of information into making a decision if this information is not coherent, contradictory or points in different directions. As a result, decision-makers may be overwhelmed by vast amounts of incoherent information and may not be able to integrate it meaningfully into their decision-making process. Again, this seems particularly likely when dealing with complex match analysis data from sports.

Another important study deals with the extent to which people can reach consensus on the basis of data (Kahan et al. 2012). Although the study from the renowned scientific journal *Nature* was on the topic of assessing the risks of climate change, this study also has relevant implications for this chapter as it describes a general psychological phenomenon: i.e., that a lack of consensus in the interpretation of data is not necessarily caused by a deficit in the understanding of data. Instead, people do not appear to interpret data objectively, but rather interpret the data in accordance with the interpretation of the data by social groups they identify with. When applied to match analysis in sports, it therefore appears possible that analysis departments may interpret match analysis data in a distorted way, for example in order not to deviate from the “anticipated opinion” of the head coach:

{...} whenever human judgment is involved, it also introduces the potential for bias {...} more information actually can make matters worse for people.

(Silver 2012, p. 92)

In summary, these findings suggest that too many options and too much information have the potential to complicate decisions (especially when people have to justify them) and leave people more dissatisfied with them afterwards. This does not necessarily mean that match analysis data cannot help with certain decisions. However, the research reviewed in this sections indicates that it is not helpful to give match analysis departments the task of collecting more and more data if there is no philosophy or theory of how to deal with this data.

QUESTIONS

What problems could arise for match analysts as a result of the research described above and what measures could be implemented to improve the ability to draw conclusions from data in sports?

- Measures to reduce the “Too-much-choice” effect
- Measures to increase the coherence of data
- Measures to reduce social impact on data interpretation

27.2 Communication of Match Analysis Data

DEFINITION

Communication

Communication is usually defined as the exchange of information or the transmission of information from a sender to a receiver (Watzlawick, Beavin and Jackson 2016). In the field of match analysis, for example, from a coach to their team. In general, information can be transferred in different forms (e.g., verbal or non-verbal) and in different ways (e.g., speaking or visualizing using tactic boards).

Communication science has identified numerous problems that arise during the transmission of information from sender to receiver. It seems obvious, for example, that communication is fundamentally hindered by cultural, developmental or intellectual differences. However, this is not a specific problem in the communication of match analysis results and will therefore not be discussed further. It should be clear to every coach that players who do not speak the coach's language must at least be supported by interpreters, or that children may need to be addressed with a different vocabulary than adult athletes. Therefore, the following sections will only review research that is of direct relevance for the communication of match analysis results; it will not go into the detail of domain-general problems in transferring information between individuals.

Studies in sports have shown that the communication of simple tactical information can influence players in such a way that certain patterns on the pitch are processed better or worse (Furley, Memmert and Heller 2010; Memmert and Furley 2007). These studies have shown that players have an attentional advantage for expected and a disadvantage for unexpected game situations. In this context, the cognitive capacity of working memory plays a central role. Working memory can be broadly defined as a mental workplace in which perceptual information and activated memory contents are combined and processed, thereby providing the necessary input for the requirements of the task at hand (Furley and Memmert 2013). Of particular relevance for the present chapter, findings from psychology show that the currently activated contents of working memory influence the orientation of the attentional focus (Knudsen 2007; Soto, Hodson, Rotshtein and Humphreys 2008). Based on these basic scientific findings, Furley and Memmert (2013) have shown that information in working memory influences attention orientation when making tactical decisions in sport. More specifically, the results show that certain decisions can be made more quickly than others. In addition, it could be observed that the quality of decisions was influenced by the activated contents of the athletes working memory. Further studies (Furley et al. 2010; Memmert and Furley 2007) show that instructions of the coach (e.g., tactical information), which should actually support the player, could have a negative impact on tactical decision-making. Applied to sports practice (especially amongst children and young players), such studies have been summarized with the slogan “Tell me less, then I see more” (Furley 2014). This rule of thumb for coaches has important implications concerning the coaching process and in fostering creative behavior in sport games (Memmert 2007, 2015).

Taken together, these studies show that communicating match analysis findings to athletes can negatively influence their behavior on the field. The information communicated influences the players’ attentional focus and can, to a certain extent, make them blind to important events on the field that have not been addressed by the coach or are not part of a particular game plan. The following anecdotal example serves to illustrate this reasoning: In an interview after the Schalke vs. Rostock game, soccer goalkeeper Manuel Neuer attempted to explain a “fatal mistake” that led to a direct goal by the opponent. Neuer said that he wanted to follow the coach’s tactical guideline to initiate quick counterattacks. Due to this tactical instruction he only focused on the potential recipient of his throw-off; he had not seen the opponent whom he (unfortunately) played the ball directly into his feet.

27.2.1 How Much Information Should Players Be Given Before a Match?

Several studies from cognitive psychology and first studies in sports illustrate that working memory plays an important role in following tactical instructions from the coach or match analyst, as it is directly related to attention control and goal-oriented behavior (Furley and Memmert 2010, 2012, 2013). A central finding in cognitive psychology is the limited capacity of working memory (Cowan 2001), especially in stressful situations. This means that players can only hold a very limited amount of information available in their working memory while they are engaged in playing the game. Therefore, coaches should not overwhelm their players with too much information in time-outs or pre-game briefings. Although there are no sport science studies that have empirically addressed the question of how much information should be conveyed to the athletes in time-outs or pre-game tactic briefings, psychological research shows that the capacity of working memory in humans probably comprises about four pieces of information (Cowan 2001). While this says little about how much or what information should be passed on to players to achieve a desired effect on the pitch, it does give a certain rule of thumb that players probably

retain relatively little information from tactic briefings, let alone use the received information on the pitch.

WARNING

It is likely, for example, that a basketball coach overloads his players during a team time-out if he first addresses the past mistakes, then the next three offensive moves, and only finally informs the team to change their defense.

27.3 Methods, Results and Discussion

In the previous section, I discussed the problems that can arise when interpreting match analysis data, deriving conclusions from it, and then communicating these to the players. Since the topic of the chapter is very general, it is not possible to identify a clear methodological approach of investigating the topic or summarizing the results. Rather, the review of cross-disciplinary findings, theories and problems is intended to show that sports science should begin to scientifically investigate how match analysis data should be used and in what form it should be communicated to athletes. As already mentioned, there is hardly any research in this area and therefore hardly any evidence-based recommendations exist. At present, a major goal of scientific efforts in match analysis attempts to correlate match analysis data with success in a variety of sport contexts. It is hoped that this branch of research will lead to the identification of so-called KPIs that can then be improved in competition and training (Low et al. 2020; Memmert and Raabe 2018). Although this is certainly important research, other questions regarding match analysis or performance analysis in general should not be neglected. Despite the lack of empirical research, the next section intends to derive rules of thumb from related research fields that can assist coaches and match analysts in interpreting and communicating match analysis data.

27.4 Practical Relevance

In science it is important to acknowledge one's own "ignorance" or limitations of research results, whereas this is probably not the case in practical fields such as match analysis in sports. In particular, coaches and match analysts are under pressure week after week and have to try to prepare their own team as well as possible for the next opponent. To ensure that the data collected from match analyses is helpful and not a hindrance in this endeavor, a few rules of thumb could be helpful.

First, it should be mandatory for match analysts in professional sports to receive a good methodical and statistical education. Further, they should be aware of the problems of data analysis and drawing conclusions from data. The previous sections of this chapter argued that match analysts probably do not help their teams if they have the attitude that as much data as possible should be passed on to coaches and teams without further reflection. Good match analysts probably differ from less good ones in terms of their ability to select and reduce large amounts of data to the most essential and important data and then communicate this accordingly to coaches or players. This sounds simple, but in fact it is far from trivial. In addition, it is highly context-dependent how a team will prepare for the next opponent and what data will be helpful in this respect. For example, teams that dominate in their respective leagues undoubtedly must prepare

differently for an opponent than teams that need every point in the relegation battle. For this reason, it does not seem to make sense to formulate recipe like rules about what information from the match analysis data should be passed on when preparing the team for the next opponent.

Nevertheless, scientific findings from well-controlled studies can be useful, for example, to show what influence even simple tactical instructions can have on players (Furley et al. 2010; Memmert and Furley 2007). From such studies, for example, the recommendation is derived that information should be communicated cautiously to players during tactical briefings (“tell me less the I will see more”). Furthermore, cognitive psychological studies show that only a very small amount of information (approximately four pieces of information) can be kept present while engaged in a task. Since these studies involved very simple information units (e.g., numbers or letters) and the carried-out tasks were short and did not last 90 minutes like a soccer game, it seems reasonable to conclude that players should not receive more than —one or two pieces of information in pre-match tactical briefings or during time-outs. Especially, since research shows that people in demanding situations (e.g., under stress or fatigue) are less receptive and their working memory capacity is further reduced (Furley and Wood 2016 for an overview).

Further practical recommendations that follow from the studies reviewed in this chapter have already been summarized in the previous Chapter 24 and will not be repeated here. As a general rule of thumb, it makes sense to optimize and standardize the process of data collection, data analysis, data transfer, and communication of results in the professional sports sector. This is certainly already well implemented in some professional areas but can probably be further improved in other areas. In this regard, clear personnel assignments, fixed time schedules and hierarchical structures are helpful in reducing large amounts of data into manageable practical recommendations. Process models of decision-making from psychology (e.g., Schwartz 2016) provide reasonable advice on how to proceed in professional match analysis in sports:

- 1) Coaches should clearly define their goals and communicate them to the match analysts. In doing so, it is important to define what is expected from match analysis data, what data should be collected and how these data will be used. Note that is common to formulate more than one goal.
- 2) The importance of the different goals should be explicitly formulated, and it should be specified which data are helpful for achieving the goals. This helps to reduce the amount of data and to concentrate on the most important data.
- 3) Coaches should critically discuss the practical implications of the data with their match analysts. They should also critically discuss how likely the conclusions drawn from the data will contribute to achieving the goals.
- 4) Finally, the coaching staff should try to reach an agreement in a critical dialogue on what is communicated to the entire team, what will possibly be communicated to parts of the team or to individual players (of course, this does not apply to individual sports but only to team sports). Furthermore, the coaching staff should decide when the tactical team meetings will take place. This can already be started during training. In this case, only a few tactical instructions might have to be repeated in the tactics briefing prior to the game to reduce the risk of players being overwhelmed by too much new information.
- 5) During the game, it should be continuously evaluated whether the team is well adjusted or whether there is room for improvement.
- 6) In the post-match period, match analysts and coaches should evaluate whether the goals have been achieved and how the process of match analysis and data-driven match preparation can be further optimized.

CONCLUSION

Scientific studies indicate that it is difficult to successfully implement large amounts of data in professional sports. Nevertheless, this chapter does not intend to criticize match analysis and “Big Data” in sport in general, but attempts to raise awareness of the problems involved. Through the coordinated cooperation of scientists, coaches and match analysts, match analysis data can be useful in informing training protocols and match preparation. The guidelines mentioned here might serve as a helpful starting point.

Questions & Answers

1. What was the central finding from the “Many Analysts”, one data set project Silberzahn et al. (2018)?
 - A. Due to their standardized methodological training, independent scientific research groups arrive at similar results when analyzing the same data set.
 - B. Numbers are interpreted uniformly and take subjectivity out of decisions in sports.
 - C. The more analysts analyze the same data set, the more reliable the result.
 - D. Independent scientific research groups arrive at different results when analyzing the same data set.
2. What problems can coaches, and match analysts face when dealing with large amounts (Big Data) of match analysis data?
 - Data are not interpreted uniformly and allow different conclusions
 - Different analysts have difficulties in finding a consensus in data
 - Coaches and match analysts can suffer from the “too much choice” effect and have difficulty integrating large amounts of information into their decision-making.
 - People have difficulties when they are confronted with contradictory data
3. How are coach addresses, working memory of the players, and behaviour on the pitch related?
 - A. There is no connection.
 - B. Coach speeches are processed in the players’ working memory. They can influence the players’ attentional focus on the pitch, giving them an advantage for expected and a disadvantage for unexpected game situations.
 - C. Tactical speeches by the trainer lead to better decision-making behaviour on the pitch via the working memory.
 - D. If the coach only addresses four points in tactical speeches, the players’ working memory will be optimally adjusted for the game
4. What are the practical consequences of the limited capacity of the working memory for trainer addresses?
 - A. None
 - B. Coaches should make full use of the working memory capacity of their players and address at least 4 points.
 - C. The players’ absorption capacity is low, especially in pressure situations, and coaches should, therefore, only address 1–2 points.
 - D. It does not matter how many points are addressed; it depends on how the issues are addressed

5. How can players in sports deal with match analysis data in a meaningful way?
 - Coaches and analysts should formulate clear goals for how the data are handled
 - Not all data are equal, and therefore data should be weighted according to their relevance
 - Communication between analysts and the coaching staff regarding the reduction of data to the most essential
 - Agreements which data will be passed on to whom
 - Critical evaluation and reflection in dealing with game analysis data

References

- Baumeister, R. (2010). *Is there anything good about men? How cultures flourish by exploiting men*. New York, NY: Oxford University Press.
- Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24, 87–114.
- Furley, P. (2014). Sag mir weniger, dann sehe ich mehr [English translation: tell me less the I will see more]. *Fußballtraining Junior*, 4, 26–79.
- Furley, P., & Memmert, D. (2010). The role of working memory in sports. *International Review of Sport and Exercise Psychology*, 3, 171–194.
- Furley, P., & Memmert, D. (2012). Working memory capacity as controlled attention in tactical decision making. *Journal of Sport and Exercise Psychology*, 34, 322–344.
- Furley, P., & Memmert, D. (2013). “Whom should I pass to?” the more options the more attentional guidance from working memory. *PLoS One*, 8(5), e62278. doi:10.1371/journal.pone.0062278
- Furley, P., & Wood, G. (2016). Working memory, attentional control, and expertise in sports: A review of current literature and directions for future research. *Journal of Applied Research in Memory and Cognition*, 5, 415–425. doi:10.1016/j.jarmac.2016.05.001
- Furley, P., Memmert, D., & Heller, C. (2010). The dark side of visual awareness in sport – Inattention blindness in a real-world basketball task. *Attention, Perception & Psychophysics*, 72, 1327–1337.
- Glöckner, A., & Betsch, T. (2012). Decisions beyond boundaries: When more information is processed faster than less. *Acta Psychologica*, 139, 532–542.
- Kahan, D.M., Peters, E., Wittlin, M., Slovic, P., Ouellette, L.L., Braman, D., & Mandel, G. (2012). The polarizing impact of science literacy and numeracy on perceived climate change risks. *Nature Climate Change*, 2, 732–735.
- Knudsen, E. (2007). Fundamental components of attention. *Annual Review of Neuroscience*, 30, 57–78.
- Low, B., Coutinho, D., Gonçalves, B., Rein, R., Memmert, D., & Sampaio, J. (2020). A systematic review of collective tactical behaviours in football using positional data. *Sports Medicine*, 50(2), 343–385.
- Memmert, D. (2007). Can creativity be improved by an attention-broadening training program? – An exploratory study focusing on Team sports. *Creativity Research Journal*, 19, 281–292.
- Memmert, D. (2015). *Teaching tactical creativity in sport: Research and practice*. Abingdon: Routledge.
- Memmert, D., & Furley, P. (2007). “I spy with my little eye!” – Breadth of attention, inattention blindness, and tactical decision making in team sports. *Journal of Sport & Exercise Psychology*, 29, 365–381.
- Memmert, D., & Raabe, D. (2018). *Data analytics in football. Positional data collection, modelling and analysis*. Abingdon: Routledge.
- Porter, T.M. (2020). *Trust in numbers: The pursuit of objectivity in science and public life*. Princeton, NJ: Princeton University Press.
- Rein, R., & Memmert, D. (2016). Big data and tactical analysis in elite soccer: Future challenges and opportunities for sports science. *SpringerPlus*, 5, Article 1410. doi:10.1186/s40064-016-3108-2
- Scheibehenne, B., Greifeneder, R., & Todd, P.M. (2009). What moderates the too-much-choice effect? *Psychology & Marketing*, 26, 229–253.
- Scheibehenne, B., Greifeneder, R., & Todd, P.M. (2010). Can there ever be too many options? A meta-analytic review of choice overload. *Journal of Consumer Research*, 37, 409–425.
- Schwartz, B. (2016). *The paradox of choice: Why more is less*. New York: HarperCollins.

- Silberzahn, R., Uhlmann, E.L., Martin, D.P., Anselmi, P., Aust, F., Awtrey, E., ... Nosek, B.A. (2018). Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science*, 1, 337–356. doi:10.1177/2515245917747646
- Silver, N. (2012). *The signal and the noise: The art and science of prediction*. London, UK: Penguin.
- Soto, D., Hodsoll, J., Rotshtein, P., & Humphreys, G.W. (2008). Automatic guidance of attention from working memory. *Trends in Cognitive Sciences*, 12, 342–348.
- Watzlawick, P., Beavin, J.H., & Jackson, D.D. (2016). *Menschliche Kommunikation. Formen, Störungen, Paradoxien*. Göttingen, D: Hogrefe.
- Wilson, D.S. (2019). *This view of life. Completing the Darwinian revolution*. New York, NY: Pantheon Books.